Java Backend Developer - Technical Assessment

Overview

Create a RESTful API for a simple shop warehouse management system using Spring Boot.

Time Allocation: 1-2 days
Submission: Share your GitHub repository link when complete

Business Context

Build a backend system for managing a shop's warehouse inventory.

Core Requirements:

- **Items**: The shop sells items that need to be tracked
- **Variants**: Items can have variants (e.g., different sizes, colors)
- **Pricing**: Each item/variant has a price
- **Stock**: Track inventory levels for items and their variants

What You Need to Deliver:

- RESTful API endpoints for managing the above
- A way to prevent selling items that are out of stock
- Basic CRUD operations
- Data persistence

Technical Constraints

Must Use:

- Java (version 17+)
- Spring Boot 3.x
- Any database you prefer (in-memory is fine)
- Git for version control

Your Choice:

- Architecture and design patterns
- API endpoint structure and naming
- Database schema design
- Additional features or validations you think are important
- Libraries and dependencies

Deliverables

1. GitHub Repository

- Clean commit history
- Include a README with:
    - How to run the application
    - Design decisions you made and why
    - Any assumptions you made
    - API endpoint examples

2. Working Application

- Should be runnable by following your README
- Include sample data (optional but helpful)

Note: There's no single "correct" solution. We're interested in your thinking process and coding style.

Good luck! We're looking forward to seeing your solution.